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| AS008IW | Sampling Procedure | |
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# Sampling Procedure for Coal(In Accordance with ISO 18283:2006)

## 1. Purpose

This procedure describes how representative samples of coal are to be obtained, reduced, and preserved for the preparation of reference materials (RMs/CRMs) or proficiency testing (PT) items. The procedure ensures that coal samples are collected in line with ISO 18283 and meet the requirements of ISO 17034 and ISO Guide 35.

## 2. Scope

This procedure applies to all coal consignments (e.g., bituminous, sub-bituminous, anthracite, lignite) sampled by the RMP for CRM production or PT schemes. It covers primary sampling, gross sample formation, sub-sampling, and reduction up to the stage where test or reference portions are obtained.

## 3. References

- ISO 18283:2006 – Hard coal and coke — Manual sampling
- ISO Guide 35 – Reference materials — General and statistical principles
- ISO 17034 – General requirements for the competence of reference material producers
- ISO/IEC 17043 – Proficiency testing — General requirements

## 4. Responsibilities

- Sampler: Executes the sampling strictly according to this SOP.
- Operations Manager and/or Laboratory Supervisor: Ensures the adequacy of sampling plans and equipment.
- Laboratory Manager: Confirms that sampling records and chain-of-custody are complete and traceable.

## 5. Definitions

- Increment: A quantity of coal collected at one point from the lot.
- Gross Sample: The aggregate of all increments from the lot.
- Sub-sample: A reduced, representative portion of the gross sample.
- Test/Reference Portion: The final portion prepared for analysis or CRM production.

## 6. Equipment

- Manual shovel or scoop (flat-edged, clean, non-contaminating material)
- Sample containers: moisture-tight, dustproof, clearly labelable
- Riffle splitter or rotary divider for sample reduction (cone and quartering)

- Calibrated balance
- Moisture-proof storage bags

## 7. Procedure

7.1 Sampling Plan
- Define lot size (e.g., 1,000 tonnes of coal).
- Determine number of increments according to ISO 18283:
  * Minimum of 35 increments for lots >100 tonnes.
  * For smaller lots, increments should be sufficient to cover the full particle size distribution.
- Document plan in the Sampling Record Form.

7.2 Collection of Increments
- Take increments from the moving stream (preferred) or a static pile (if unavoidable).
- Each increment shall be at least 3 times the nominal top size (NTS) of the coal particles.
- Collect increments at evenly distributed time/space intervals to ensure representativity.
- Avoid bias (e.g., do not select only fine or coarse material).

7.3 Formation of Gross Sample
- Combine all increments into a single gross sample.
- Mix thoroughly to minimize segregation.
- Weigh and record the total mass of the gross sample.

7.4 Sample Reduction
- Reduce the gross sample to a manageable size using riffle splitters or rotary dividers (never by coning and quartering unless validated).
- Continue reduction until the sub-sample is:
  * Large enough for homogeneity and stability testing
  * Suitable for CRM production or PT distribution
- Record reduction steps, including masses before and after each reduction.

7.5 Sample Handling & Preservation
- Transfer sub-samples into clean, moisture-tight containers.
- Label each container with:
  * Sample ID
  * Date
- Store samples in a cool, dry environment to prevent moisture loss and oxidation.

## 8. Quality Control

- Collect duplicate gross samples for reproducibility checks.
- Retain a reference sample (archive) for traceability.
- Record ambient conditions (temperature, humidity) at the time of sampling.
- Verify homogeneity of reduced samples by testing subsamples.

## 9. Safety

- Wear PPE (dust mask/respirator, gloves, goggles, safety boots).
- Minimize exposure to coal dust; use dust suppression where possible.
- Follow site safety rules for working near stockpiles and conveyors.

## 10. Records

- Sampling Record Form (lot details, increment plan, sampler signature).
- Gross and reduced sample weights.
- Chain-of-custody forms.
- QC check records.
- Equipment calibration logs.